



BRAC UNIVERSITY
CSE360: Computer Interfacing
Lab Project Report

**Title: Smart Door Security and Anti-Tailgating System Using
Arduino UNO and ESP32-S3 CAM**

by

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Abstract

Tailgating this being the case where an unauthorized individual passes through a door right behind someone who has properly unlocked it—overcomes most conventional access control systems since these systems check a credential rather than keeping count of the actual number of people who go through. To overcome this shortcoming, this project designs a prototype for door security that combines credential verification with physical occupancy counting. The Arduino UNO R3 serves as the main controller and is responsible for handling an MFRC522 RFID reader via SPI, an SSD1306 OLED via I2C, a pair of FC-51 infrared beam sensors, a magnetic reed switch, an active buzzer, and a relay which operates a 12 V solenoid lock. Meanwhile, an ESP32-S3 camera module runs alongside it as a separate visual channel, sending video data over WiFi and sending a signal to the Arduino using a dedicated open-drain hardware line.

The system operates according to the order in which the beams are triggered. Two infrared sensors are placed a short distance apart, and the sequence in which they are interrupted enables the controller to determine not only that a person has passed but also the direction in which they were moving. Each authorised unlock should therefore result in exactly one inbound crossing; if a single credential causes two crossings then this is regarded as tailgating and the lock immediately re-engages. Moreover, the system is able to tell the difference between events detected by the beams alone, events detected by the camera alone, and cases in which both systems detect the event within a configurable time interval, promoting the last-mentioned case to a confirmed alarm.

Bench testing was used to test the function of each subsystem individually and in combination. The relay and solenoid worked well when instructed to do so by the software, the reed contact was able to determine the door, the beam pair was able to count the direction of the crossings and the Arduino was able to receive the alert signal from the ESP32. During the building process several hardware faults were found and sorted out, such as a level-shifter which was found not to be suitable for use with SPI, a relay module that caused the door to unlock during a microcontroller reset, and intermittent jumper wiring that made a working RFID module seem dead. Full details of these problems and the way they were diagnosed are given, as they ended up forming a major portion of the engineering work. The system consumes less than 300 mA when idle and makes use of modules that are widely available and can be replaced separately, thus keeping both its operating cost and its repair cost low.

Keywords

Embedded systems, access control, anti-tailgating, sensor fusion, microcontroller interfacing, SPI, I2C

1. Introduction

In offices, laboratories and residential buildings electronic access control has mostly taken the place of mechanical keys. All that is required is to show a card, a fob or give a biometric scan, after which the system checks whether the credential is valid and then unlocks the door. The system is convenient and can be kept record of, and it is effective at dealing with the threat for which it was intended—that of someone who has no credential at all.

It is much less effective at preventing tailgating; if an attacker just waits by the door and walks through after an authorised user, no credentials are presented and the access logs show only one legitimate entry. The door has no way of knowing that two people passed through it. This is not a hypothetical vulnerability. Physical security evaluations regularly include tailgating as one of the simplest and most dependable methods of gaining entry to a restricted area, exactly since it takes advantage of ordinary human politeness rather than relying on any technical fault.

The reason for the problem is that a normal reader will respond to the question "is this credential valid?" but will never consider the one asking "how many people actually walked through?" These are distinct questions and in order to answer the second one a sensor is needed which monitors the doorway rather than the card.

1.1 Objectives

The aim of this project was to create a working prototype which would answer both of the questions. The specific objectives were:

- Use an MFRC522 module to implement RFID-based credential verification via SPI with an Arduino UNO R3.
- Use a pair of infrared beam sensors to detect how many people go through the doorway and in which direction by looking at the order in which the beams are interrupted.
- Issue a clear tailgating alarm when the number of people passing is greater than the number of authorisations granted.
- Include an independent visual channel by using an ESP32-S3 camera module which is connected to the Arduino via a dedicated hardware signal line not through network messaging.
- Mix the evidence from the beam and the camera so that any events seen by both sensors are raised above those seen by only one.
- Operate a genuine 12 V solenoid lock via a relay, ensuring proper electrical isolation between the logic and load sections.
- Continuously report the system's status on an OLED display and by means of distinct audible alarm patterns.

1.2 Scope and Relevance

The prototype is not fitted into a door frame but instead is built on a desk, and the sensors are arranged so that they will be activated by a person's hand rather than by someone walking by. Although the detection logic is the same in both situations, the beam sensors chosen have enough range to be mounted on a real door frame without any need for software changes.

The project is directly related to the concerns of a course on computer interfacing since it involves three different digital communication protocols working at the same time on a single microcontroller, deliberately.

The management of a system that includes both 3.3 V and 5 V levels, the isolation between a low-voltage logic domain and a high-current inductive load, the inter-processor signalling between two boards which are independently powered, and the real-time processing of asynchronous sensor

events without causing a blockage all involved practical difficulties, which are described later in this report.

2. Related Work and Inspiration

RFID door lock projects are very common in both hobbyist and academic circles. The usual way of implementing them is to use an MFRC522 reader with an Arduino, compare the card's UID against a list that has been stored, and then operate a relay or a servo to unlock the door. Many published tutorials and student projects follow this approach in almost the same way. While these designs are helpful for learning about SPI communication and relay control, all of them have the same flaw that was mentioned in the introduction: the system keeps a count of card taps rather than of people.

There are commercial remedies for preventing tailgating. Optical turnstiles and mantrap doors physically allow only one person to pass at a time, and more advanced setups make use of overhead stereo cameras or time-of-flight sensor arrays to determine the number of people present. Although they work well they are expensive and are generally only installed in high-security locations. Because of the difference in cost between a basic RFID reader and a complete optical turnstile, most normal doors have no protection against tailgating.

There is also a considerable amount of work carried out by hobbyists using ESP32 camera modules, the majority of which involves streaming video to a browser or performing light image classification. These projects usually regard the camera as a standalone device that communicates via WiFi.

2.1 What Differentiates This Project

The project is intentionally positioned between the two extremes. Instead of using an expensive occupancy sensor, it employs two cheap infrared beam modules and obtains a people count from the timing relationship between them. The order in which the two beams are broken provides both a count and a direction, which is enough information to identify the particular failure mode of interest.

The other key decision relates to how the camera is integrated. In most ESP32 camera projects the devices communicate via WiFi, which causes network latency, means that both devices have to be on the same network, and results in a dependence on infrastructure that might fail. In this case, however, the ESP32 sends a signal to the Arduino using a single GPIO line set up as open-drain, so the alarm is transmitted in microseconds and keeps working even if the network is not available. While the camera stream is still available through WiFi for the purpose of evidence and monitoring, the alarm system does not rely on it.

Third, the system does not regard its two sensing channels as being interchangeable. It notes which sensors have observed an event and increases the level of response when they agree. If an intruder is detected solely by the beams, this triggers one alarm level; if the same event is confirmed by the camera, a higher alarm level results. This represents a simple instance of sensor fusion and implies that in order to defeat the system it is necessary to break both channels rather than just one.

3. Technical Approach

3.1 System Architecture

The system is based on two microcontrollers, each with clearly defined roles. The Arduino UNO R3 functions as the controller responsible for making decisions. It controls all the sensors which monitor the physical door, assesses the credentials, operates the state machine, and is the only device allowed to trigger the lock. The ESP32-S3 camera module acts as a separate sensing peripheral. It carries out video capture and streaming, and the only link it has to the decision-making process is a single digital line which it can pull low in order to generate an alert.

This was intentionally done. By having all the safety-critical decisions taken on a single microcontroller, the lock cannot be operated as a result of a network event, and the Arduino is able to carry out its state machine correctly even if the ESP32 crashes, loses its WiFi connection, or is completely disconnected. The two boards are connected except for a common ground and one signal wire.

The architecture can be divided into four layers. The input layer includes the RFID reader, the two infrared beam sensors, the reed switch, the manual override input, and the camera alarm signal. The processing layer is made up of the Arduino firmware, which is organised as a finite state machine with six states. The output layer consists of the OLED display, the buzzer, and the relay that operates the solenoid. The power layer ensures that the 5 V logic domain and the 12 V lock domain are electrically separated, with connection between them being made only through the relay contacts.

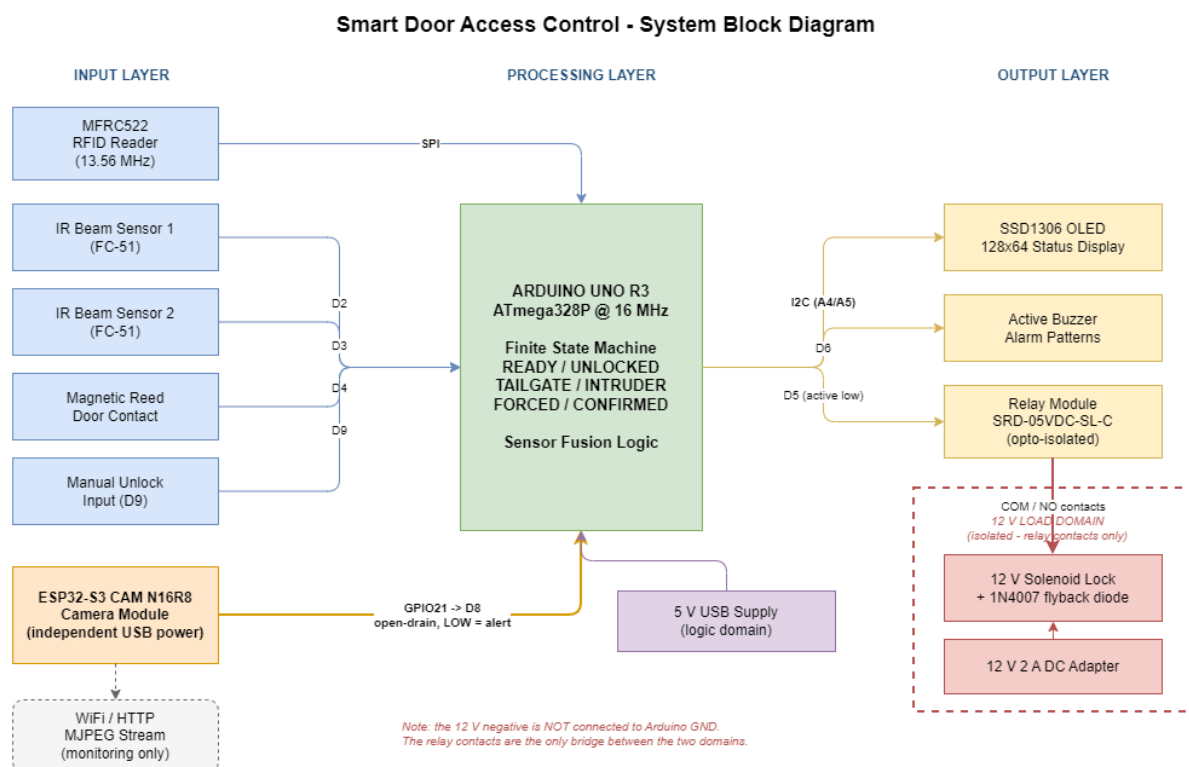


Figure 1: System block diagram showing the four functional layers and the isolation boundary between the logic and load domains.

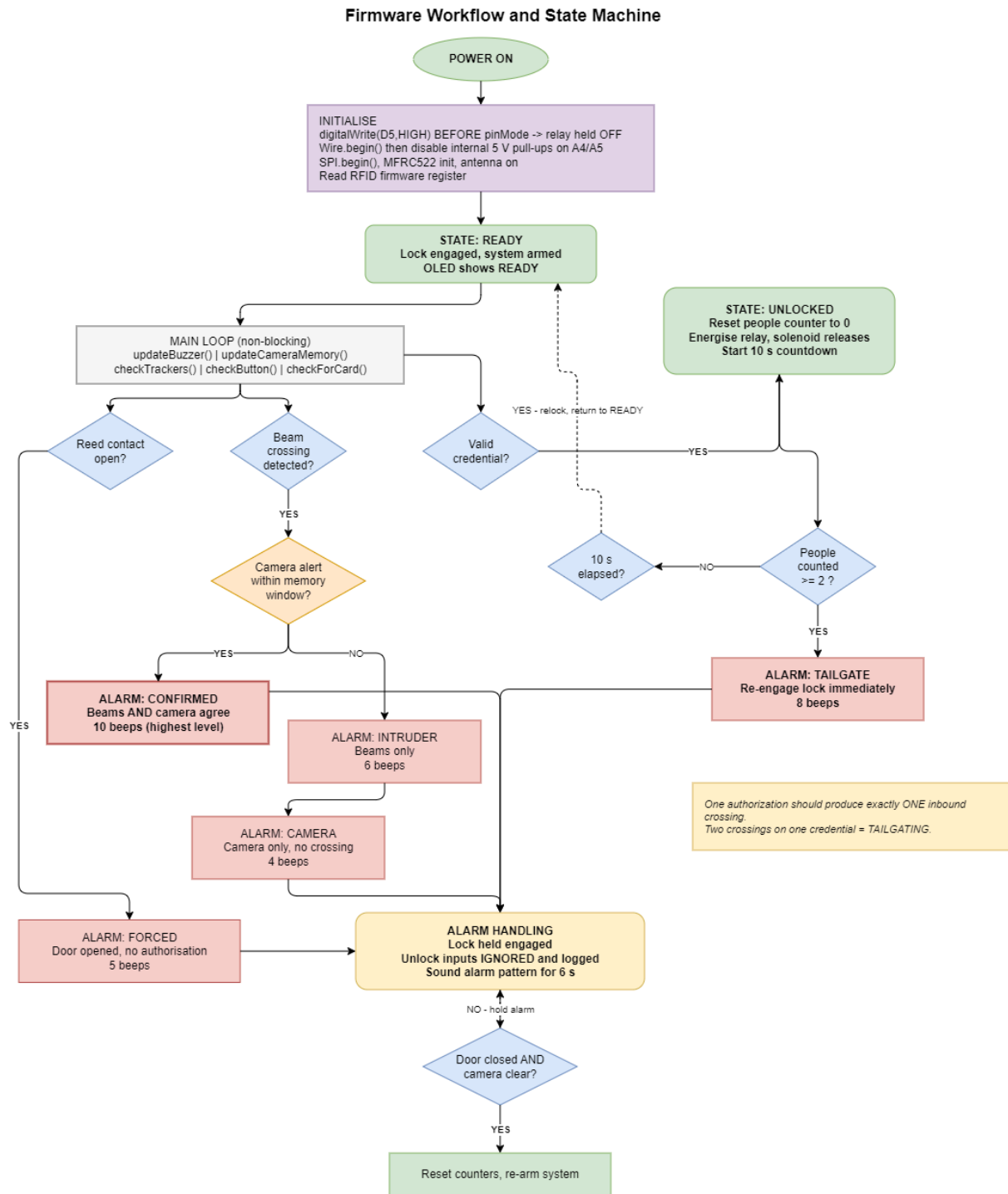


Figure 2: Firmware workflow, showing the state machine and the decision path that combines beam and camera evidence.

3.2 Components Used

The hardware was chosen because it was available in the local market and since any individual module could be replaced without the need to redesign the rest of the system.

Component	Specification	Function in the System
Arduino UNO R3	ATmega328P, 16 MHz, 32 KB flash, 2 KB SRAM	Main controller; runs the state machine and drives all outputs
ESP32-S3 CAM N16R8 (RBD-2765)	ESP32-S3-WROOM, 16 MB flash, 8 MB OPI PSRAM, OV-series camera	Video capture and streaming; raises the camera alert line
MFRC522 RFID reader	13.56 MHz, ISO/IEC 14443A, 3.3 V, SPI	Credential verification
FC-51 IR sensor module (x2)	LM393 comparator, adjustable 2–30 cm, 5 V, digital output	Directional people counting
Magnetic reed contact	Two-wire normally-open switch	Door open/closed state; forced entry detection
SSD1306 OLED display	0.96 in, 128x64, I2C, address 0x3C	Live system status
Relay module	SRD-05VDC-SL-C, opto-isolated, 10 A @ 30 VDC contacts	Switches the 12 V lock circuit
Solenoid door lock	12 V DC, fail-secure	Physical locking mechanism
DC power adapter	12 V, 2 A	Powers the lock circuit
1N4007 diode	1000 V, 1 A rectifier	Flyback protection across the solenoid coil
Active buzzer	5 V, two-pin	Audible alarm patterns
Passive components	Resistors, DC barrel jack, jumper wires, breadboard	Interconnection and signal conditioning

Table 1: the various hardware components and the role of each in the system.

With regard to the software, the development was done using the Arduino IDE (version 2.x). For RFID communication the Arduino firmware makes use of the MFRC522 library from GithubCommunity, while for the display it uses the Adafruit SSD1306 and Adafruit GFX libraries, and for the two buses the built-in SPI and Wire libraries are employed. The ESP32 firmware is based on the Espressif CameraWebServer example, with modifications made to include the alert line and a small secondary HTTP server on port 8080 for the purpose of manually triggering an alert during testing.

3.3 Cost Breakdown

The prices given are only approximate retail prices in Bangladeshi Taka at the time of buying and must be regarded as indicative rather than precise.

Item	Qty	Unit (BDT)	Subtotal (BDT)
Arduino UNO R3	1	950	950
ESP32-S3 CAM N16R8 (RBD-2765)	1	2,200	2,200
MFRC522 RFID module with card and fob	1	250	250
FC-51 infrared sensor module	2	90	180
Magnetic reed door contact	1	120	120

Item	Qty	Unit (BDT)	Subtotal (BDT)
0.96 in I2C OLED display	1	450	450
1-channel 5 V relay module	1	130	130
12 V solenoid door lock	1	800	800
12 V 2 A DC adapter	1	450	450
DC barrel jack with leads	1	30	30
1N4007 diode	1	5	5
Active buzzer	1	25	25
Resistor assortment	1	20	20
Breadboard and jumper wires	1	400	400
Total (approximate)			6,010

Table 2: An approximate breakdown of the costs.

To put it in context, a simple commercial RFID door controller costs the same amount but provides no tailgating detection, whereas an optical turnstile able to carry out the same function is one to two orders of magnitude more expensive. The worth of this design is that it includes occupancy verification at a cost equivalent to that of two infrared modules.

3.4 Communication Protocols

There are four different communication methods in the system, each one having been selected for reasons that are specific to the device it is used on.

The MFRC522 RFID reader makes use of SPI. In order to carry out card authentication, the reader needs a synchronous, full-duplex, multi-byte exchange, a requirement which SPI meets at high speed with a simple hardware setup. On the UNO the hardware SPI facility is connected to pins D11 (MOSI), D12 (MISO) and D13 (SCK), with D10 serving as the slave select. SPI was chosen over the alternative I2C and UART modes offered by the MFRC522 since the I2C bus was already being used by the display and because SPI's dedicated slave-select line keeps the reader separate from the other traffic.

The SSD1306 OLED uses I2C, as it transmits a complete 1024-byte frame buffer and is not timing-critical, and since I2C only needs two wires it saves up the UNO's limited number of pins. The bus operates at 400 kHz in order to decrease the amount of time the microcontroller is blocked waiting for the display updates. The display responds to the address 0x3C.

A dedicated GPIO line is used to send alerts from the ESP32-S3 to the Arduino, and this connection needs to be explained in detail since a simple implementation is not reliable. The ESP32 produces 3.3 V logic, whereas the ATmega328P, which is operating at 5 V, needs an input of more than 0.6 times VCC, that is over 3.0 V, to recognise a logic high. Although a 0.3 V margin is technically within specification, it is easily reduced by the resistance of the wiring and by electrical noise. Instead of incorporating a level-shifting component, the signal polarity was reversed; the ESP32 pin is set up as an OUTPUT_OPEN_DRAIN and the Arduino pin as an INPUT_PULLUP, so that the line remains high when left alone thanks to the 5 V pull-up on the Arduino side and the ESP32 lowers it to ground

when it wants to send an alert. Pulling the signal low is clear and does not require any voltage translation, and this setup also ensures that no 5 V potential is ever applied to the ESP32 pin.

The video stream is transmitted from the ESP32-S3 to any browser within the same network via WiFi and HTTP, using the conventional multipart MJPEG streaming method. On port 8080 there is a second lightweight HTTP server which provides a manual alert trigger for use during testing and demonstrations. Crucially, the alarm system does not rely on this network connection; WiFi is used solely for carrying the evidence and monitoring.

Protocol	Devices	Pins / Channel	Justification
SPI	Arduino <-> MFRC522	D10 SS, D11 MOSI, D12 MISO, D13 SCK	Fast synchronous multi-byte exchange required for card authentication
I2C (TWI)	Arduino <-> SSD1306	A4 SDA, A5 SCL	Two-wire bus conserves pins; display is not timing-critical
GPIO (open-drain)	ESP32-S3 -> Arduino	GPIO21 -> D8	Microsecond latency, no network dependency, no level shifter needed
WiFi / HTTP	ESP32-S3 <-> browser	802.11 b/g/n, ports 80 and 8080	Video streaming and remote monitoring

Table 3: Communication protocols and the reasoning behind each choice.

3.5 Circuit and Pin Assignment

The pin assignments were decided only after sorting out a number of conflicts; D0 and D1 were left free all the way through because they are used for the hardware serial port which is employed for programming and debugging, and having a load on these pins would prevent the code from being uploaded. The hardware SPI peripheral reserves pins D10 to D13, and the only pins on the UNO that are capable of supporting hardware I2C are A4 and A5, so neither of the two groups could be assigned to other functions.

Arduino Pin	Connected To	Mode	Notes
D0, D1	Unused	-	Reserved for hardware serial; must remain clear
D2	IR sensor 1 output	INPUT	Also INT0, available for interrupt upgrade
D3	IR sensor 2 output	INPUT	Also INT1
D4	Reed contact	INPUT_PULLUP	Second reed wire to GND; LOW = door closed
D5	Relay module IN	OUTPUT	Active low; forced high before pinMode
D6	Buzzer positive	OUTPUT	Driven in short pulses only
D8	ESP32-S3 GPIO21	INPUT_PULLUP	LOW = camera alert

Arduino Pin	Connected To	Mode	Notes
D9	Manual unlock input	INPUT_PULLUP	Shorted to GND to trigger
D10	MFRC522 SDA/SS	OUTPUT	Hardware SPI slave select
D11	MFRC522 MOSI	OUTPUT	Hardware SPI
D12	MFRC522 MISO	INPUT	Hardware SPI; 3.3 V input read directly
D13	MFRC522 SCK	OUTPUT	Hardware SPI
A4	OLED SDA	I2C	Internal pull-ups disabled in firmware
A5	OLED SCL	I2C	Internal pull-ups disabled in firmware
5 V	IR sensors, OLED, relay, buzzer	Power	Approximately 250 mA total
3.3 V	MFRC522 VCC and RST	Power	RST tied high to keep the reader awake

Table 4: Final Arduino pin assignment.

On the 12 V side the positive terminal of the adapter is connected to the COM terminal of the relay and the normally-open contact is connected to the positive of the solenoid. The negative of the solenoid is then connected back to the negative of the adapter. The normally-closed contact is not used, so when the relay is de-energised the lock remains unpowered and therefore has fail-secure behaviour. A 1N4007 diode is fitted directly across the solenoid terminals with its

Placing the cathode on the positive terminal gives a route for the collapsing magnetic field when the coil is switched off and thus stops the resulting voltage spike damaging the relay contacts.

The Arduino's 12 V negative terminal is on purpose left disconnected from the Arduino ground. The relay contacts are the only link between the two systems. This arrangement keeps the solenoid's switching transients from entering the logic ground and is the standard approach when switching an inductive load in conjunction with sensitive digital circuitry.

See Figure 3: Circuit Schematic- Pin Connections and Power Domain Isolation. This figure shows the circuit schematic with all the connections and the isolation boundary between the 5 V logic domain and the 12 V load domain.

Circuit Schematic - Pin Connections and Power Domain Isolation

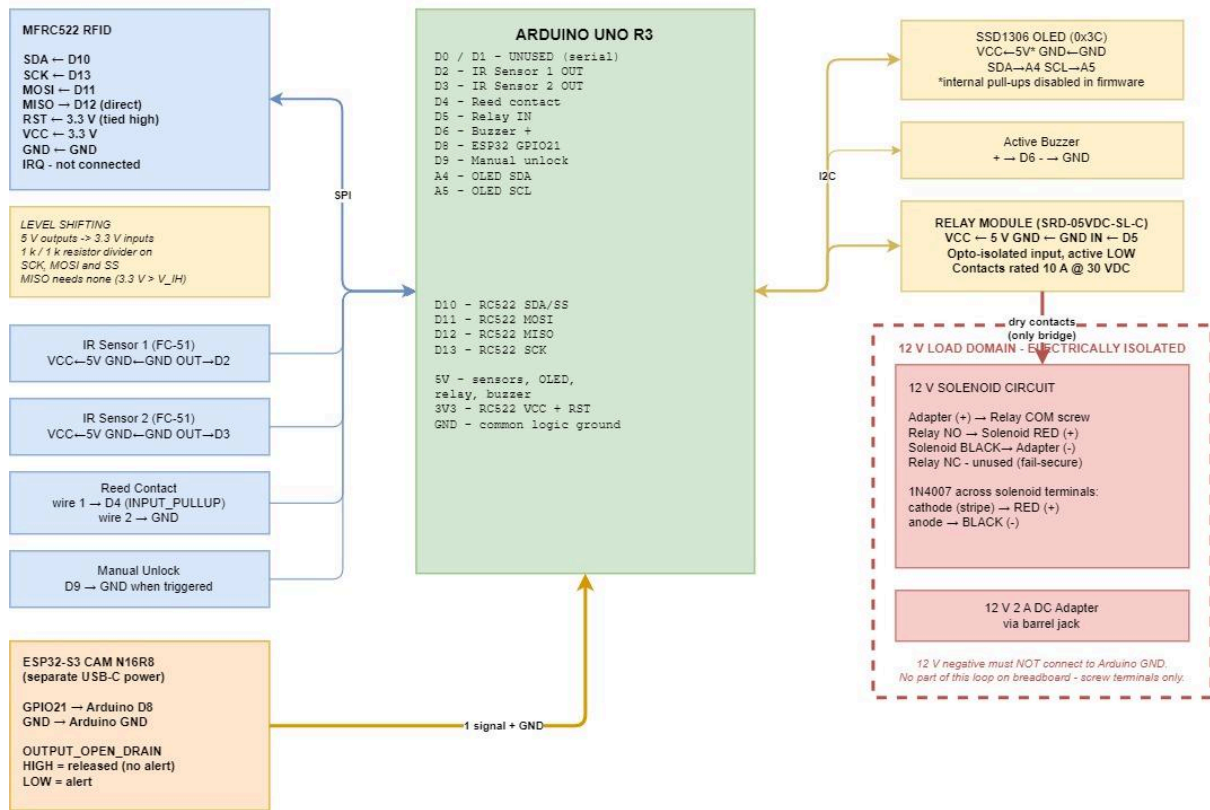


Figure 3: Circuit schematic showing all connections and the isolation boundary between the 5 V logic domain and the 12 V load domain.

3.6 Implementation

The firmware is arranged in the form of a finite state machine, having six states which are assessed once during each loop iteration. These states are READY (which means armed and locked), UNLOCKED (meaning that authorised access has been granted), TAILGATE, INTRUDER, FORCED, and CONFIRMED. Each of the alarm states keeps the lock engaged.

The loop is designed to be completely non-blocking; there are no instances of `delay()` in the main loop since such a blocking call would cause the system to miss the beam crossings. The buzzer patterns, the alarm timeouts, and the unlock countdown are all based on comparisons with `millis()`. The display is only refreshed when the content that is visible changes rather than being updated on each iteration, because a complete transfer of a 128x64 frame over I2C takes about 25 ms and carrying out the transfer continuously would slow the loop down enough to cause fast events to be missed.

3.6.2 The logic for access control and tailgating

The UID of a card is checked against a list which has been stored. If there is a match, the people counter is set back to zero, the relay causes the solenoid to be energised, and the system then goes into the UNLOCKED state for ten seconds. The main rule is clear in that each authorisation must result in exactly one inbound crossing. Should the counter reach two while the system is in the UNLOCKED

state, the firmware immediately reactivates the lock, gives eight beeps, and causes the system to enter the TAILGATE state.

When a crossing is detected while the system is in READY status, indicating that no authorisation had been granted at all, the INTRUDER state results. Also, if the reed contact shows that the door is opening while the system is in READY, the FORCED state is entered.

3.6.2 Access Control and Tailgating Logic

When a card is presented, its UID is compared against a stored list. On a match, the people counter is reset to zero, the relay energises the solenoid, and the system enters the UNLOCKED state for ten seconds. The core rule follows directly: one authorisation should produce exactly one inbound crossing. If the counter reaches two while the system is in UNLOCKED, the firmware immediately re-engages the lock, sounds eight beeps, and enters the TAILGATE state.

A crossing detected while the system is in READY, meaning no authorisation was granted at all, produces the INTRUDER state instead. Independently, if the reed contact reports the door opening while the system is in READY, the FORCED state is entered.

3.6.3 Sensor Fusion

The camera and beam channels are combined by means of a short memory window. The firmware keeps track of the timestamp of the latest camera alert and, upon a beam crossing, checks if a camera alert occurred within that window. This is necessary since the two channels do not trigger at the same millisecond; the camera picks up motion when a person is approaching, while the beams are interrupted a little later as they are passed.

Beams	Camera	Resulting State	Alarm Pattern
Crossing detected	Alert within window	CONFIRMED	10 beeps
Crossing detected	No alert	INTRUDER	6 beeps
No crossing	Alert active	CAMERA	4 beeps
Two crossings on one unlock	Either	TAILGATE	8 beeps
Any	Any (reed opens while armed)	FORCED	5 beeps

Table 5: Decision matrix combining the beam and camera channels.

The value of this setup lies in the fact that an attacker has to overcome two completely different kinds of sensing systems if they are to go undetected; even if the infrared beams are blocked the camera will still function, and if the camera is avoided the infrared beams will still be there.

3.6.4 Fail-Safe Design

Two features of the firmware deal with failure modes that are easy to miss when considering security. The first of these relates to the relay at startup. The relay module in question is active-low, and when the microcontroller is reset all the pins are set up as high-impedance inputs. In that time if the relay

input floats or is pulled low, the door will unlock. To prevent this the firmware sets `digitalWrite(RELAY_PIN, HIGH)` before it calls `pinMode(RELAY_PIN, OUTPUT)`, thereby activating the internal pull-up while the pin is still in input mode and keeping the relay in its inactive state during the changeover. If the two calls were made in the opposite order, which is the more usual practice, the door would click open each time during a reset and each time a new code is uploaded.

The second measure ensures that alarms cannot be silenced by means of signal. So long as an alarm condition as an alarm condition is active, the manual unlock and card input is, in fact, a recorded act. An intrusion alarm that can be dismissed can be dismissed by the same action that would have prevented it.

3.7 Challenges Encountered

A large part of the time devoted to the project was used for diagnosing hardware problems rather than actually writing code, the faults being recorded here in the order in which they were met, since the diagnostic process was a major component of the work.

3.7.1 The bidirectional level converter is not suitable for use with SPI

The first design employed a four-channel BSS138 bidirectional level converter in order to reduce the Arduino's 5 V SPI outputs to the 3.3 V that was needed by the MFRC522. However, when the reader was switched on it returned a

The firmware version register read 0x00, indicating no response; lowering the SPI clock to about 400 kHz made no difference.

The problem is due to a feature of the BSS138 topology, not a wiring mistake. Each channel actively lowers its line via a MOSFET but is able to raise it only through a 10-kilo-ohm resistor. With approximately 50 pF of capacitance from the breadboard and the wires, the time constant that results is about 500 ns, so it takes approximately 1.1 microseconds for a rising edge to reach a valid logic high. At a frequency of 400 kHz, the half-period of the SPI clock is 1.25 microseconds, which means the rising edge occupies almost the entire half-period. Since, in SPI mode 0, the data is sampled on the rising edge, the sampled data was invalid. The BSS138 board is designed for use with I2C, which is naturally open-drain and can tolerate this behavior; SPI is push-pull and cannot.

The converter was taken out and replaced with resistor dividers consisting of 1-kilo-ohm and 1-kilo-ohm pairs, which resulted in a Thevenin source impedance of about 500 ohms and a time constant of of around 25 ns, that is, roughly 20 times faster.

3.7.2 Intermittent Jumper Wiring Masquerading as a Dead Module

Even after the level converter had been removed, the readings were still inconsistent and kept alternating between 0x00 and 0xFF with successive readings. If a stable wrong value had been obtained, this would have indicated a dead device; but since the value changed from one reading to another, it showed that the input was floating and picking up electrical noise, which pointed to the MISO line.

The problem was not solved by a systematic check of the power supply, the pin arrangement, or continuity. Finally, the reader was reconnected by means of completely new jumper wires, after which it gave a steady 0x90 and kept doing so afterwards. It was later discovered that several of the jumper wires in the set were intermittent or open. This took up several hours and is the most important practical lesson from the project: faults in the connectors simulate symptoms that are very similar to those caused by a faulty component, and a diagnostic test should therefore be carried out using known-good wiring before deciding that a module is dead.

3.7.3 An unidentified firmware version identifier

When communication had been set up, the firmware still showed the reader as not working. The validation logic only accepted the officially documented MFRC522 version identifiers 0x91, 0x92, 0x88 and 0xB2, while the module actually in use reported 0x90, a code that is used by certain clone devices. The check was therefore expanded to also include this value. It is a useful reminder that clone modules do not always conform to the published register values and that validation logic should be designed to fail in an informative way.

3.7.4 Connection of Power to the Relay Module

At first the relay did not operate at all, with no audible click and no indicator LED lighting up. The Arduino control logic had been confirmed as correct in the software. The problem was found to be due to the relay module's VCC connection having been inserted into the wrong header position, which left the module without power. After the error had been corrected, the relay worked immediately. It is worth noting the diagnostic value of the module's own power LED: by checking it first, the fault could have been identified in just a few seconds.

3.7.5 The bouncing of the input on the manual unlock signal line

The manual unlock input, which was achieved by briefly touching a jumper wire to ground, recorded six separate access grants from a single contact. During a single touch, mechanical contacts are operated many times. A software debounce was then included, this requiring a clean transition from low to high, together with a 1.5-second lockout period during which further transitions were ignored.

3.7.6 The over-counting caused by the infrared beams

The FC-51 modules first indicated a count of eight to fifteen people when a single hand was passed over them. As explained in section 3.6.1, this happens because the narrow infrared beam is separately interrupted by the fingers, the gaps, the wrist and the arm. In order to address this, a 40 ms settle time and a person cooldown period were introduced. The counting mechanism has been adjusted to work with a moving hand in the demonstration setup; a complete human body, however, presents a larger and more continuous obstruction and would therefore require a shorter cooldown value.

3.7.7 The sensor range is not adequate for a doorway.

The first design employed reflective tracker modules that worked properly during testing but had an effective range of only 2 to 30 millimetres. This range is suitable for line-following robots but is

completely unsuitable for detecting a person going through a doorway. These modules were then replaced by FC-51 modules, which provide an adjustable range of 2 to 30 centimetres. Since both types of device have the same three-wire interface and give a digital output, no changes to the firmware were needed at all, thus confirming that the choice to keep the sensor interface generic was the right one.

3.7.8 An unintended 5 V voltage on the I2C bus

The SSD1306 controller is a 3.3 V device and when `Wire.begin()` is called on an AVR the internal pull-up resistors on the SDA and SCL pins are enabled, connecting them to 5 V. The display seemed to work normally, but its data pins were kept above their maximum rated voltage each time the bus was idle. If two `digitalWrite` operations follow immediately after `Wire.begin()` then these pull-ups are disabled and the correctly referenced 3.3 V pull-ups belonging to the display module are left to hold the bus. This defect is not visible during operation and could probably reduce the display's service life.

3.7.9 Microcontroller Reset Loop

Towards the end of the assembly process the Arduino started to restart continuously, displaying only its first two startup messages before resetting again. The fact that this behaviour occurs points to a collapse of the power supply rather than to a software fault. When all the peripherals had been disconnected and then reconnected in groups with a reset carried out after each group, the problem was found to be due to the wiring that had been disturbed during a pin reassignment. The method of reconnecting the components incrementally turned out to be much more efficient than examining the entire circuit at once.

3.7.10 The high-current path on the breadboard

At first the 12 V solenoid circuit was taken through the breadboard and connected with jumper wires. The spring contacts on the breadboard are only rated for about 1 A and have a contact resistance that grows as the contacts are switched repeatedly, whereas the solenoid draws more than 1 A with a higher inrush current. This situation therefore involves both a reliability risk and a thermal hazard. The whole of the 12 V loop has since been transferred to the relay's screw terminals using thicker gauge wire and joints that were either soldered or taped, so that no section of the high-current path remains on the breadboard.

4. Sustainability and Impact

4.1 Sustainability

The logical part of the system uses about 250 to 300 mA at 5 volts, that is less than 1.5 watts, and this level of consumption is continuous. The solenoid is the main load at more than 1 A, but it only operates during the unlock period, which lasts ten seconds for each authorised entry. If a door is used fifty times a day, the solenoid will be active for well under ten minutes within a 24-hour period, so the average power consumption is mainly due to the logic rather than the actuator.

The 5 V rail might be lowered even more; the two infrared modules together draw about 50 mA and, if they are switched on only when the PIR sensor detects someone approaching, idle power consumption could be greatly reduced. The OLED could in a similar way be turned off when no one is present. However, neither of these changes has been incorporated into the present prototype although both are simple to implement.

The design chooses discrete, separately replaceable modules instead of using one single integrated board. In the case when the RFID reader was thought to be faulty, the diagnostic procedure called for replacing just one cheap component rather than throwing away the entire unit. This level of granularity helps to extend the product's service life and reduces electronic waste because a fault in one subsystem does not render the rest useless. All the components including the relay, the sensors, the display and the reader are standard items which can be obtained independently and repaired or replaced at a low cost.

The Arduino, the ESP32 and all the sensor modules have been taken from earlier coursework and will still be reusable afterwards, and the breadboard construction is completely non-destructive. The only consumable used is the jumper wire set, since a number of these wires failed during the project and had to be discarded. Better quality connectors would result in less waste and less debugging time.

4.2 Impact

At present, tailgating protection is available only in the case of organisations that can afford optical turnstiles or mantrap portals. The high cost of the equipment and of installing it means that university departments, small offices, laboratories, server rooms and residential buildings—all of which are subject to the same threat—cannot access it. This project shows that it is possible to incorporate meaningful detection into an existing access control installation at a cost equivalent to that of two infrared sensor modules.

The detection logic also provides useful information in addition to generating alarms. Since the beam pair keeps a record of direction, the system is able to determine how many people are in the space at any given time. This is of direct relevance to fire safety and to emergency evacuation, where it is genuinely important to know the number of people in a area, and also applies to laboratory access records since the number of people present may be important for safety compliance.

From an educational point of view the project combines three communication protocols, mixed-voltage interfacing, inductive load switching, inter-processor signalling, and non-blocking real-time programming into one system. The debugging record in section 3.7 is of independent value as it provides a documented instance of systematic hardware fault isolation.

4.3 Future Work

- Instead use a time-of-flight sensor array or an overhead stereo camera, in that way the system will be able to tell the difference between two people walking together and a single person who is carrying a large object, unlike the present beam-order method.
- Transfer the function of counting votes from the polling method to hardware interrupts on pins D2 and D3. These pins already have the capability of generating external interrupts,

which would remove any possibility of missing a fast crossing when the loop is busy with a display update.

- Add logging of local events to either an SD card or an EEPROM, including timestamps, so that there will be an auditable record of access independent of any network connection.
- On the ESP32-S3 implement person detection directly on the device, taking advantage of its sufficient PSRAM to run a lightweight detection model and thus replacing the present frame-difference method with actual person counting from the camera.
- Include a supervised alarm loop with an end-of-line resistor connected to the reed contact, in order that a tamper event may be detected if the sensor wiring is cut or shorted.
- Move the circuit from the breadboard to a soldered perfboard or a custom PCB, thus getting rid of the connector reliability problems which had taken up most of the debugging effort.
- Make sure that a battery backup is added to the logic power supply so that the system can continue to monitor and log during a power cut.

4.4 Limitations

The limitations have been identified and are clearly stated rather than ignored.

The microcontroller, the ATmega328P, has 2 KB of SRAM, and this amount is entirely used up by the SSD1306 frame buffer alone. Because of this there is only a small amount of spare space and this puts a limit on how much data the firmware can hold; if a larger display or extra buffering were needed then a different microcontroller would be required.

The beam-order counting method is unable to tell the difference between a person and an object of the same size since it deduces the presence of people from obstructions; it also fails to distinguish between two people walking side by side from them walking one behind the other; for it to work properly the two beams must be crossed in a clear sequence.

The infrared beam cooling down is a compromise; if the time is set too short then one person will be counted more than once whilst if it is set too long two people entering in quick succession will be counted as a single person. The present setting has been adjusted for the demonstration arrangement not for actual pedestrian traffic and would therefore require re-tuning if the door were to be installed.

The demonstration setup included a manual unlock feature as well as the RFID reader, thus reducing its reliance on the reader while the reader's intermittent wiring fault was being diagnosed. In a system that has been put into operation, this manual input would be eliminated or substituted with a supervised request-to-exit button.

The camera alarm at the moment is either activated manually or is triggered by a heuristic based on frame size, the heuristic determining a scene change from the JPEG compression size. This is a rough stand-in for motion detection and reacts just as quickly to changes in lighting as it does to people. Although it does show the signalling path and the fusion logic accurately, it is not a person detector.

In the end, the prototype is mounted on a breadboard and offers no environmental protection; it has not been tested for electromagnetic interference, temperature variations, or the mechanical vibrations that a door frame undergoes during ordinary use.

5. Results and Discussion

The testing was done bit by bit. The various subsystems were checked one at a time before being combined, after which the entire system was run through a set of defined scenarios. The results were recorded from the Arduino's serial output at a speed of 9600 baud and from the OLED display.

5.1 Subsystem Verification

Subsystem	Test Performed	Result
OLED display	I2C address scan and text rendering	Detected at 0x3C; renders correctly
MFRC522 reader	Firmware version register read	Returns 0x90 consistently; SPI link functional
IR sensor pair	Sequential beam interruption	Inbound and outbound crossings correctly distinguished
Reed contact	Magnet removal and replacement	FORCED raised on removal; cleared on replacement
Relay module	Software toggle of D5	Audible actuation; indicator LED follows state
Solenoid lock	Direct 12 V application, then via relay	Actuates under both conditions
Buzzer	Pattern generation	Distinct patterns produced per alarm type
ESP32 camera	Browser stream over WiFi	Live MJPEG stream confirmed
ESP32 alert line	Manual trigger, GPIO21 to D8	Received by Arduino; CAMERA state entered

Table 6: Subsystem verification results.

5.2 Scenario Testing

Five operational scenarios were carried out, with each one being repeated several times and serial output being recorded in each case.

Scenario	Action	Expected Outcome	Observed
Authorised entry	Credential presented, one crossing	UNLOCKED, one person counted, relock after timeout	As expected
Tailgating	Credential presented, two crossings	TAILGATE, immediate relock, 8 beeps	As expected
Intrusion	Crossing with no credential	INTRUDER, 6 beeps	As expected
Forced door	Reed opened while armed	FORCED, 5 beeps	As expected
Camera alert	ESP32 alert with no crossing	CAMERA, 4 beeps	As expected

Scenario	Action	Expected Outcome	Observed
Confirmed event	Camera alert and crossing within window	CONFIRMED, 10 beeps	Logic verified; both channels confirmed independently
Alarm lockout	Unlock attempted during alarm	Input ignored and logged	As expected

Table 7: Scenario test results.

Representative serial output from the intrusion and camera-only scenarios is reproduced below.

```
ALARM: camera alert (no beam crossing)
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```
*** TAILGATING DETECTED *** IR beams: 2 people on ONE authorised entry
```

```
ALARM: intruder (IR beams only, no camera)
```

5.3 Discussion

In relation to the objectives stated in section 1.1, the system achieves its main aims. All the required functions—credential verification, directional people counting, detection of tailgating, integration of the cameras by hardware line, solenoid actuation, and status reporting—have been carried out and verified.

The tailgating detection works correctly at the logical level since two crossings on a single authorisation consistently result in the TAILGATE state and an immediate relock. Although the accuracy of the count underneath is constrained by the sensing method, beam-order counting is merely an indication of people rather than a direct measurement, and the need to adjust the settings in order to prevent multiple counts from a single pass shows the compromise that has to be made. While the system is better at responding to the question "did more than one obstruction pass?" than answering "exactly how many people passed?", for the particular threat of tailgating the first of these questions is the one that matters.

The individual branches of the sensor fusion logic were verified. In the case of events based only on the beam and those based only on the camera, each one produced the right distinct states and alarm patterns, which showed that both channels are working and that the decision logic is able to discriminate between them. The reliable attainment of the combined CONFIRMED state during the demonstration was found to rely on the two triggers occurring within the memory window, and since the window duration can be adjusted, it is a tunable parameter.

The biggest result in terms of the amount of work put in did not come about during the design phase but during the physical building of the device. Three different faults were found to be due to problems with the connectors rather than to faulty components or software errors, and these accounted for most of the time spent debugging. Although modern electronic modules are usually quite reliable, the method of assembling the circuit using a breadboard and jumper wires—which enables rapid prototyping—is also the least reliable part of the system. This clearly indicates that soldered construction should be adopted for any actual deployment, and the lesson applies very well to projects other than this one.

It is also important to point out that some of the defects identified during construction were latent rather than obvious at first sight; the 5 V pull-ups used on the 3.3 V display and the relay unlocking during microcontroller reset both resulted in systems that seemed to function correctly. Neither of these faults could have been detected solely by functional testing; they were both discovered through reasoning about the behaviour stated in the datasheet regarding what the circuit was actually doing.

6. Conclusion

The aim of this project was to deal with a particular and well-known weakness in traditional electronic access control, namely its inability to detect tailgating – that is, an unauthorised individual passing through a door behind someone who has properly unlocked it – since these systems check credentials rather than keeping a count of the number of people.

This prototype enhances a conventional RFID door controller by means of two cheap infrared beam sensors for detecting occupancy. The system works out not only that a person has passed through but also in which direction they moved, and then compares this number with the number of authorisations that have been granted. As a second, separate sensing option, an ESP32-S3 camera module is connected to the main controller via a dedicated open-drain signal line rather than through the network, so that the sending of alerts does not rely on WiFi being available. If both channels have reported an event within a short time period, the system raises it to a confirmed alarm.

All the main objectives have been achieved; the system checks the credentials via SPI, counts the directional crossings, picks up cases of tailgating and immediately reengages the lock, detects forced entry by means of a magnetic reed switch, accepts and responds to camera alarms, operates a real 12 V solenoid lock via a properly isolated relay circuit, and reports its status on an OLED display using distinct audible signals for each alarm category.

A large portion of the work was devoted to diagnosing hardware faults rather than to the design or coding phase. It was discovered that a bidirectional level converter was fundamentally inappropriate for use with SPI because of its resistive pull-up configuration; intermittent jumper wires caused symptoms which could not be told apart from those of a defective module; a relay was found to open the door at the time of microcontroller reset; and the I2C library was found to be supplying 5 V to a 3.3 V display. All of these points are given in section 3.7. Two of them were latent faults which functional testing could never have detected.

From the point of view of sustainability, the design is modest in terms of power, consuming less than 1.5 W continuously and only switching on the solenoid for a short period each time someone enters, and since it uses separate, replaceable modules, the system does not have to be discarded if one component fails. The greater significance of this approach is in terms of cost: at present tailgating detection is a requirement for equipment that is beyond the means of most small institutions, and this study shows that it is possible to include useful detection at a cost equivalent to that of two sensor modules. The occupancy figure produced by the system also has further uses in relation to emergency evacuations and for complying with laboratory safety regulations.

The limitations are genuine and are described in section 4.4: beam-order counting works by deducing the presence of people from obstacles and is unable to tell when two people are walking side by side, the cooldown tuning represents a middle ground between over-counting and under-counting, and the

camera-side motion detection is based on a heuristic not a person detector. Overcoming these issues would demand better sensing hardware rather than better logic, and the ways in which this can be achieved are given in section 4.3. Although the architecture is sound, it could take on those improvements without having to be redesigned.

7. Contribution

Name	ID	Responsibilities	Specific Contributions
Sawda Nawar Pink	22201668	Hardware Design & Circuit Assembly	- Selected components and checked voltage compatibility between the 3.3 V and 5 V parts - Built the breadboard circuit and assigned pins around the fixed SPI, I2C and serial pins - Designed the resistor divider level shifting for the RC522 SPI lines - Verified all connections and traced faults down to individual jumper wires
Ahmad Al Jaber	22201110	Arduino Firmware & Camera Integration	- Designed the six-state machine and wrote the directional people-counting logic - Implemented RFID over SPI and OLED over I2C, disabling the AVR's 5 V pull-ups to protect the display - Added the relay fail-safe ordering so the lock cannot release during reset - Configured the ESP32 camera server and the open-drain alert line to Arduino D8 - Wrote the sensor fusion logic combining beam and camera evidence
Masudur Sadik Rifat	22201145	Power, Actuation & Electrical Safety	- Wired the 12 V solenoid loop through the relay screw terminals, off the breadboard - Fitted the 1N4007 flyback diode with the cathode on the solenoid positive - Kept the 5 V logic and 12 V load domains isolated, bridged only by the relay contacts - Diagnosed the dead relay to a misplaced VCC lead using the module's power LED - Calculated the 5 V current budget against the UNO's supply limit

Mahir Muntasir Rafsan	22201104	Testing, Diagnostics & Documentation	• Ran subsystem verification and the seven scenario tests, capturing serial logs as evidence • Diagnosed the BSS138 failure on SPI from its rise time against the clock period • Traced the intermittent RFID fault to faulty wiring rather than a dead module • Isolated the reset loop by reconnecting peripherals in groups • Prepared the report, diagrams and cost analysis
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